Marten Weijer

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About

I am a master's student in Data Science at Minnesota State University, Mankato, advised by Prof. Rushit Dave. I'm drawn to questions about how AI is changing software development, and I want to use my PhD to find where I can contribute most. My interests include AI for software engineering, empirical software engineering, and software analytics. I am applying to PhD programs for Fall 2027.

Research

The Missing Middle: A Systematic Review of Machine Learning for Municipal-Level Residential Energy Forecasting
M. Weijer, R. Dave, M. Bhavsar
2026 IEEE International Conference on Emerging Trends in Networks and Computer Communications (ETNCC), Windhoek, Namibia, 2026

Interpretable Machine Learning for Municipal Residential Gas Consumption: Integrating Climate, Building Stock, and Socioeconomic Drivers
M. Weijer, R. Dave, M. Bhavsar
Submitted to *Sustainability* (MDPI), Special Issue on Artificial Intelligence, Image Processing, and Data-Driven Solutions for Sustainable Development, 2026

Education

Minnesota State University — Mankato, MN
Master of Science in Data Science — *Jan. 2026 – May 2027*

Amsterdam University — Amsterdam, The Netherlands
Bachelor of Science in Computer Science — *Aug. 2013 – May 2017*

Experience

Graduate Teaching Assistant — Aug. 2026 – Present
Minnesota State University – CIS 483: Web Applications and User Interface Design — *Mankato, MN*

- Grade assignments, hold lab office hours, and give code-review feedback on student web applications

Senior Software Engineer — Feb. 2021 – Jan. 2026
Booking.com — *Amsterdam, The Netherlands*

- Designed and maintained large-scale data systems, with a focus on testing, code review, CI and reliable delivery across teams
- Built and operated backend services and event-driven pipelines in Java, Spring and Kafka, deployed on Kubernetes
- Led design discussions and code reviews, and mentored engineers on testing practices and maintainable code
- Took part in on-call rotations, debugging production incidents and following up with root-cause analysis

Software Engineer — May 2017 – Feb. 2021
Capgemini B.V. — *Amsterdam, The Netherlands*

- Developed Python and SQL pipelines and automated test suites, working in version-controlled, CI-driven development workflows
- Built and maintained backend applications and REST APIs for enterprise clients across several projects
- Worked in Scrum teams, turning client requirements into technical designs and shipping features in regular sprints

Certifications

Kafka: Confluent Certified Developer for Apache Kafka (CCDAK) — 2024
SAFe: Certified SAFe Practitioner — 2024
Spring: Spring Certified Professional for Java Developers — 2024
Kubernetes: Certified Kubernetes Application Developer (CKAD) — 2023
Scrum: Professional Scrum Master I — 2023
Java: Oracle Certified Java SE Developer — 2022